

Ad Pricing Strategy Comparison

Per-second linear model vs. spot/play-based model — assumptions, formulas and worked scenarios

1. Overview

Two pricing structures were developed for booking taxi-top screen inventory. Both price on the same underlying factors — screens booked, route/zone, and ad playtime — but they treat playtime differently, which changes how bookings scale in cost as ad length and frequency increase.

Model A prices strictly by raw seconds of playtime held per screen per day. Model B prices per loop rotation ("play"/"spot"), with a modest multiplier for slot length rather than a direct multiplication by seconds. Model B is the model currently agreed on for development.

2. Model A — Per-Second Linear Pricing

Rate = Base rate (KES/second) × seconds of ad length × zone multiplier × volume multiplier, applied per screen per day.

Formula components

- Base rate: illustrative KES 15 / second / screen / day
- Zone multiplier: Estate 1.0x · CBD 1.5x · Airport corridor 2.0x
- Volume discount: 1–9 screens 1.0x · 10–49 screens 0.9x · 50+ screens 0.8x
- Advertiser selects: ad length in seconds, and a booking duration in days

Worked examples

Scenario	Calculation	Result
1 screen, CBD, 5s ad, 1 day	$15 \times 5 \times 1.5$	KES 112.50/day
1 screen, CBD, 20s ad, 1 day	$15 \times 20 \times 1.5$	KES 450/day
1 screen, CBD, 60s ad, 1 day	$15 \times 60 \times 1.5$	KES 1,350/day
20 screens, estate, 20s ad, 14 days	$15 \times 20 \times 1.0 \times 0.9 \times 20 \text{ screens} \times 14 \text{ days}$	KES 75,600 total
100 screens, mixed zone (1.25x), 60s ad, 30 days	$15 \times 60 \times 1.25 \times 0.8 \times 100 \text{ screens} \times 30 \text{ days}$	KES 2,700,000/month

Strengths

- Simple to explain in one sentence: pay per second held, per screen, per day
- Directly proportional — easy to justify to a finance/procurement buyer

Weaknesses

- Assumes value scales linearly with duration, but on a moving taxi-top most exposure is glance-length regardless of ad length
- Punishes long + frequent bookings multiplicatively, which can price out larger fleet clients
- No natural cap tied to actual loop capacity — a few long ads can crowd out the rotation

- Base rate (KES 15/second) is illustrative only; not benchmarked against real cost or competitor data

3. Model B — Spot/Play-Based Pricing (agreed direction)

Rate = Base price per play at a reference slot length $\times$ slot-length multiplier $\times$ zone multiplier $\times$ volume multiplier, per screen. Advertiser buys a number of plays per day rather than a duration in seconds.

Formula components

- Base price per play: illustrative KES 8 / play / screen, at a 10s reference slot
- Slot-length multiplier (sub-linear): 10s 1.0x $\cdot$ 15s 1.3x $\cdot$ 30s 1.8x $\cdot$ 60s 2.5x
- Zone multiplier: Estate 1.0x $\cdot$ CBD 1.5x $\cdot$ Airport corridor 2.0x
- Volume discount: 1–9 screens 1.0x $\cdot$ 10–49 screens 0.9x $\cdot$ 50+ screens 0.8x
- Advertiser selects: slot length (fixed options), and number of plays/day

Worked examples

Scenario	Calculation	Result
1 screen, CBD, 15s ad, 20 plays/day, 14 days	$10.40 \times 1.5 \times 20 \text{ plays} \times 14 \text{ days}$	KES 4,368 total
20 screens, estate, 15s ad, 20 plays/day, 14 days	$10.40 \times 1.0 \times 0.9 \times 20 \text{ plays} \times 20 \text{ screens} \times 14 \text{ days}$	KES 52,416 total
100 screens, mixed zone (1.25x), 60s ad, 15 plays/day, 30 days	$20 \times 1.25 \times 0.8 \times 15 \text{ plays} \times 100 \text{ screens} \times 30 \text{ days}$	KES 900,000/month

Strengths

- Matches how DOOH/radio/TV inventory is conventionally sold — advertisers already understand "plays per day"
- Sub-linear slot-length multiplier reflects that attention doesn't scale 1:1 with ad duration
- Naturally supports a loop-capacity cap: total sellable plays/day per screen is finite, so pricing and inventory management use the same unit
- More advertiser-friendly at scale — long/frequent fleet bookings don't spike in cost the way they do under Model A

Weaknesses / open items

- Base price per play (KES 8) is still illustrative — needs to be reverse-engineered from real loop length, operating hours, and total sellable plays per screen per day to be defensible
- Requires fixed slot-length options rather than fully custom ad lengths, a minor constraint on creative flexibility

4. Side-by-Side Comparison

Factor	Model A — Per-second	Model B — Spot/play (agreed)
Pricing unit	Raw seconds held	Loop plays (spots) per day
Length scaling	Strictly linear (60s = 12 $\times$ a 5s ad)	Sub-linear (60s $\approx$ 2.5 $\times$ a 10s ad)
Frequency control	Implicit (duration in days only)	Explicit lever (plays/day)
Loop-capacity fit	No natural cap	Maps directly to sellable inventory

Factor	Model A — Per-second	Model B — Spot/play (agreed)
Cost at scale (100 screens example)	≈ KES 2.7M/month	≈ KES 900K/month
Industry familiarity	Lower — bespoke unit	Higher — matches DOOH/broadcast convention

5. Recommendation & Next Step

Model B (spot/play-based) is the agreed direction. The remaining open item is replacing the illustrative KES 8 base price with a figure derived from actual loop length, operating hours per day, and total sellable plays per screen — so the base rate is scarcity-driven rather than assumed.